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# PROBABILITY & STATISTICS

# BEG203SH

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# CHAPTER ONE

## INTRODUCTION AND DISCRIPTIVE STATISTIC

**1.1. Statistics:** The word statistic is derived from german word "statistik" which mean the political state. Statistics is the science which deals with the collection, presentation (Classification, tabulation, diagram and graphs), analysis and interpretation of the numerical data

**Function/Application of statistics**

The major functions of statistics are elaborated in subsequent sections.

a) **Statistics avoids collecting data by avoiding our biases:** Data are the fundamental elements of statisticsand they need to be collected. Collecting data sounds simple but may be the most difficult, particularly when we want to avoid biases in the process of collecting data. There are numerous scientific sampling methods in statistics which avoids biases while collecting data. Findings based on data, which are free ofbiases, can be used to estimate parameters and such estimate get public credibility.

b) **Statistics provides techniques for organizing data scientifically**: Collected data are in the raw form. Theyneed to organize scientifically. Statistics provides techniques that help us to organize data scientifically. Nowadays it is customary to organize data in computer. Organization or management of data is very essentials in modern research. since well organized data eases the work of data analysis as well as helps fordrawing desired information very quickly.

c) **Statistics enables to draw findings from masses of data:** One of the main objectives of data analysis is to draw desired findings from masses of data. There are statistical methods or techniques that help us to draw desired findings from a large data set. The desired findings can be used in variety of situations, such as comparison of the performance between two or more business companies, formulating policies, and to take decisions.

d) **Statistics helps to take decision under uncertainty:** One major topic of inferential statistics is hypothesis testing, which enables us to take decision under uncertainty with desired amount of risk.

e) **Statistics enables to interpret data**: The derived findings from a data set may need to interpret so that the people of various walks of life can understand the meaning of the findings. Statistics provides knowledge for interpreting the findings. vi. Statistics enhances presentation of findings effectively Presentation of numerical facts to public is as important as finding of facts from the data. Statistics provides various methods, such as tabular. graphical, summary measures, statistical models etc. for presenting the numerical findings.

f) **Statistics enhances presentation of findings effectively**: Presentation of numerical facts to public is as important as finding affect from the data. Statistics provides various methods, such as tabular. graphical, summary measures, statistical models etc. for presenting the numerical findings.

**Limitation of statistic**

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Like in many other disciplines, statistics has certain limitations. Some of them are listed below.

i. **Statistics works on aggregate level only** : Statistical findings are usually interpreted in terms of average, which may not be true to an individual. For example, statistical finding, such as, average monthly salary of employees of a company is Rs 40,000, but no employee in the company may be receiving this amount of salary.

ii. **Statistics is relatively weak in the analysis of categorical data**: Categorical data are frequently occurred in practice. However, there are some statistical methods that do not work for categorical data, but they work Of quantitative data.

iii. **Statistics is likely to be misused and imsinterpreted:** Statistical methods are often intentionally misused, in the sense finding ways to interpret data that are favorable to the presenter. Inappropriate use of the available tools can produce misleading conclusions. In this context, it is worth mentioning the famous quote believed to be due to Benjamin Disraeli: There are three types of lies - lies, damn lies, and statistics.

iv. **Statistics cannot prove anything**: The logic employed in inferential statistics is inductive in nature (drawing inference from small part - sample - to a larger part - population) which is opposite to the


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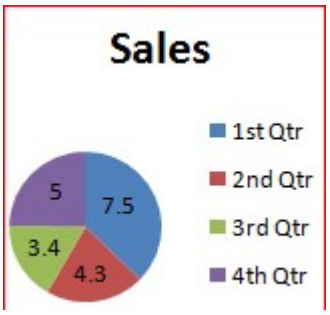
deductive logic used in mathematics. The deductive logic or argument does not prove anything.

v. **Statistical results are sometimes distrusting:** Sometimes conflicting nature of statistical statements are available in the literature, particularly in the medical sciences. For example, statements like "doing X reduces high blood pressure" and also statements like "doing X actually worsens high blood pressure". This kind of conflicting statement is mainly when studies were conducted on different groups with different protocols. However, many readers may fail to notice these distinctions, or the media may oversimplify this vital contextual information and the public's distrust of statistics is thereby increased.

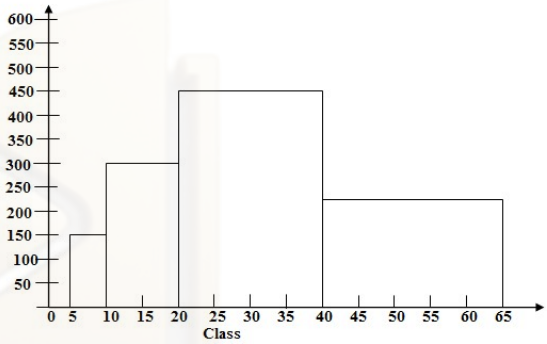
vi. **Statistic is only mean and its laws are not exact**
vii. **It does not measure a qualitative phenomena which cannot be measure in quantity.**

**Pictorial Representation of Data**

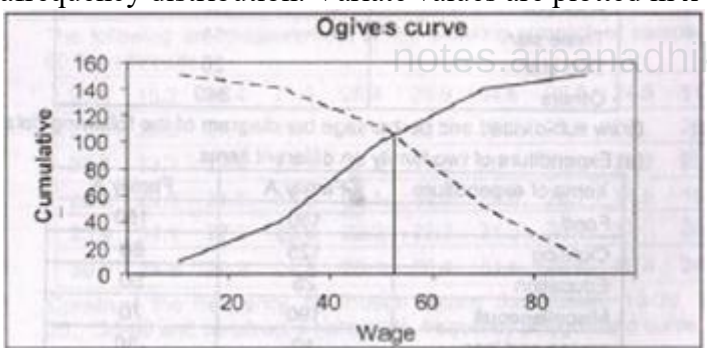
a) **Pie chart:** A pie-char is a circular diagram which is usually used for depicting the components of a single factor. Pie chart is an angular two dimensional diagram in which area of different sectors of circle can represent items of statistical data. The circle is divided into segments, which are in proportion to the size of the components. To calculate the angle of each component, we first equate the total observation equal to 360° i.e. angle at center of the circle. Then the corresponding angle or any given value $= \frac{360°}{\text{Total value}} \times$ given value




b) **Histogram**: A Histogram is a graph which is suitable for frequency distribution with continuous classes, the width of all bars is equal to class interval and heights of the bars are in proportion to the frequencies of the respective classes For unequal class intervals the heights will be proportional to the frequency density. It is ratio of frequency to corresponding class size.
∴ Frequency density= $\frac{\textbf{class frequency}}{\textbf{length of class width}}$




c) **Ogive curve:** It is a graph plotted for the variate values and their corresponding cumulative frequency of afrequency distribution. Variate values are plotted in x-axis and cumulated frequency in y- axis




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## 1.3.Measure of center tendency/ MEASURE OF LOCATION

The measure of central tendency is defined as an average. It is a single value within which the range of data tends to cluster which represents a groups of individual values in a simple and concise manner so that the mind can get a quick understanding of the general idea of the individual in the group. Since the value lies within the range of data, it is known as measure of central tendency. The **objectives of averages** are

i. To facilitate comparison
ii. To know about population from a sample.
iii. To present the silent features of mass of complex data.
iv. To trace statistical relations.



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v. To helps in decision making.

**Properties/Characteristics of Good/ideal Measure of Central Tendency**

The desire properties or requisites of a good measure of central tendencies are as follows.

a. It should be based on all the observations of a set of values.
b. It should be rigidly defined and its values should be definite.
c. It should be easily understand and computable.
d. It should be least affected by extreme values.
e. It should fluctuate least from sample to sample drawn from the same population.
f. It should be suitable for further statistical and mathematical treatments

**TYPES OF CENTRAL TENDENCIES**

1. **Arithmetic Average/Arithmetic mean**: Arithmetic mean of set of observation may be defined as the sum ofobservation divided by the total number of value in the set. It is denoted by $\bar{x}$

   i.e. $\bar{x} = \frac{\text{Sum of observation}}{\text{Number of observation}}$

   $\therefore \bar{x} = \frac{\sum x}{n}$

   **Calculation of mean**

   **a. For Individual series**

   Let $x_1, x_2, X_3 \ldots X_n$ be the values of the variables .then mean is defined as

   $\bar{x} = \frac{\sum x}{n}$

   **b. For Discrete Series and continuous series**

   For discrete and continuous data also known as frequency data, Let $X_1, X_2, X_3, \ldots\ldots\ldots\ldots X_n$ be n observation with corresponding frequency $f_1, f_2, \ldots\ldots\ldots\ldots f_n$. Then the arithmetic mean or simply mean is defined as:

   $\bar{x} = \frac{\sum fx}{N}$

   Where $N = \sum f$ is total frequency

2. **Median :**Median is a positional average i.e. its value depends on the position occupied by a value in the frequency distribution. Median is the value of the variable that divides the ordered set of values in to two equal halves i. e. 50% value are to the left of median and 50% are to the right of median . . It is most preferable measure of location for asymmetric distribution.

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**Calculation of median**

**A. For individual series**

a) Arrange the data in either ascending or descending order of magnitude.
b) If the number of observations is odd , the middle value gives the median and if number of observations is even there will be two middle values so the arithmetic mean of two middle values gives the, median.
c) Formula for calculating the median is as,

Median = Value of $\left(\frac{n+1}{2}\right)^{th}$ item.

Here n is the number of observations.

**B. For discrete series**

i. Arrange the data according to their ascending order of magnitude,
ii. Form the cumulative frequency distribution.
iii. To calculate median we use the following formula as

Median = Value of $\left(\frac{N+1}{2}\right)^{th}$ items,

Where 'N' is the total num.ber of observation.


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**B. For Continuous series**

a) Prepare the less than cumulative frequency distribution
b) Find the value of N/2
c) The class with cumulative frequency just equal or greater than $\frac{N}{2}$ is median class and this contains median.
d) The value of median is calculated by using following formula.

Median = $L + \frac{\frac{N}{2} - c.f}{f} \times h$

Where L is lower limit of median class,
N = ∑f, total frequency,
C. f. is cumulative frequency proceeding to the median class, 'f' is the frequency of median class, and 'h' is size of median class

3. **Mode:** The mode of the values of variable in a series of data is that value of a series which appear most frequently than any other and around which the other items of the set concentrate densely. It is denoted by $M_o$.

**Computation of mode in case of continuous series**

For frequency distribution with classes in ascending order of magnitude, the class corresponding to the maximum frequency is called model class and we use the following formula for calculation of mode.

$$M_o = l + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

Where L is lower limit of model class
$\Delta_1 = f_1 - f_0$ and $\Delta_2 = f_2 - f_0$
$f_1$ is maximum frequency .
$f_o$ is frequency preceding to model class
$f_z$ is frequency succeeding to model class

4. **Partition Values**

Partition values are variate values of variable, say X, which divide the total number of observations into equal number of parts. The equal number of parts may be of four, eight. ten, twelve, one hundred etc.

**TYPES OF PARTITION VALUES**

1. **Quartiles :** Three variate values of the variable, say X, which divide the series or frequency in to four equal parts, are called quartiles for the corresponding distribution of X. There are three quartiles namely $Q_1$ , $Q_2$, $Q_3$.

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**Calculation of quartiles**

**a. For individual:**

$Q_i$ = value of $[i\,(\frac{n+1}{4})]^{th}$ item where i=1,2,3

**b. For Discrete:**

$Q_i$ = value of $[i\,(\frac{N+1}{4})]^{th}$ item where i=1,2,3

**c. For continuous:**

$Q_i = L + (\frac{i\frac{N}{4} - c.f}{f}) \times h$ where i=1,2,3

i. L is lower limit of the class in which the partition value lies.
ii. cf is the cumulative frequency preceding to the class in which the partition values lie.
iii. h is size of class or class width.
iv. F is the frequency of the class in which the partition value lies.

2. **Deciles :** Nine variate values of the variable, say X, which divide the series or frequency into ten equal parts, are called Deciles for the corresponding distribution of X. There are nine Deciles namely $D_1$ , $D_2$, $D_3$.



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. .. . $D_9$

**Calculation of Deciles**

**a. For individual:**

$Q_i$ = value of $[i\,(\frac{n+1}{10})]^{th}$ item where i=1,2,3 .....9

**b. For Discrete:**

$Q_i$ = value of $[i\,(\frac{N+1}{10})]^{th}$ item where i=1,2,3.....9

**c.For continuous:**

$Q_i = L + (\frac{\frac{N}{4}-c.f}{10})$ x h where i=1,2,3 .....9

Where ,

i. L is lower limit of the class in which the partition value lies.
ii. cf is the cumulative frequency preceding to the class in which the partition values lie.
iii. h is size of class or class width.
iv. F is the frequency of the class in which the partition value lies.

**3.Percentiles :** Nintynine variate values of the variable, say X, which divide the series or frequency into hundred equal parts, are called Deciles for the corresponding distribution of X. There are nine Deciles namely $P_1$ , $P_2$, $P_3$.....$P_{100}$

**Calculation of percentiles**

**a. For individual:**

$Q_i$ = value of $[i\,(\frac{n+1}{100})]^{th}$ item where i=1,2,3 .....99

**b. For Discrete:**

$Q_i$ = value of $[i\,(\frac{N+1}{100})]^{th}$ item where i=1,2,3.....99

**c.For continuous:**

$Q_i = L + (\frac{\frac{N}{4}-c.f}{100})$ x h where i=1,2,3 .....99Where ,

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i. L is lower limit of the class in which the partition value lies.
ii. cf is the cumulative frequency preceding to the class in which the partition values lie.
iii. h is size of class or class width.
iv. F is the frequency of the class in which the partition value lies

**Dispersion:** Dispersion is defined as the scatterness of the items from the central values of data. It is also called **the measurement of variability** in the given set of data.

## METHODS OF MEASURE OF DISPERSION

**A. Range:** It is simplest and easiest method of studying the variation in the set of observations. It is defined as the difference between the largest and smallest values of a set of data. If 'L' is largest and 'S'· is the smallest value in the data then,

**Range=L – S**

**Co-efficient of range:** Relative measure of range called coefficient of Range is defined as Coefficient of Range (C.O.F)

C.O.F= $\frac{L-S}{L+S}$


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**B. Quartile Deviation:** The difference between the upper and lower quartiles is called inter quartile range; symbolically it is equal to ($Q_3$ -$Q_1$). Half the inter quartiles range is known as semi- inter quartile range or quartile deviation (OD) and given as;

**Q.D**$=\frac{Q3-Q1}{2}$

**Co-efficient of Quartile deviation :**The relative measure of dispersion based on quartiles also known as coefficient of QD is defined as

**Co-efficient of Quartile deviation**$=\frac{Q3-Q1}{Q_3+Q_1}$

**Merits of Quartile deviation**

It is easy to calculate and understand.

ii. It can be calculated in case of open end frequency distribution as well.

iii. It is not affected by 25% upper and 25% lower extreme values.

**Demerits of Quartile deviation**

i. It is based only on the central 50% observation and ignores the other 50%.

ii. It is not applicable to further algebraic treatment.

iii. ·It is not based on all the observations.

iv. It is affected by fluctuation of sampling.

**C. Mean deviation:** Mean deviation is a measure of dispersion which is based on all the values of set of data. It is defined as the average of the absolute deviation taken from an average, usually, the mean, median or mode. It is denoted by MD.

**Merits of Mean Deviation**

i. It utilizes all the observations of the series.

ii. It is simple to calculate and understand

iii. It is least affected by extreme values.

**Demerits of Mean deviation**

i. The foremost weakness of mean deviation is that in its calculation negative differences are considered as positive without any sound reasoning

ii. It is not an amenable to further statistical treatment.

iii. It can't be calculated in case of open end frequency distribution if mean is used as average.

**Calculation of mean deviation**

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a. **For individual: MD**$=\frac{\sum|X-A|}{n}$ where A may be any chosen constant out of mean , median and mode.

b. **For Discrete and continuous: MD**$=\frac{\sum f|X-A|}{N}$ where A may be any chosen constant out of mean , median and mode; N=$\sum$f is total frequency

**Co-efficient of MD:** $\frac{\text{Mean deviation}}{\text{average used}}$

**D. Standard deviation:** Standard deviation is the absolute measure of dispersion. It fulfills all the requisites of a good measure of dispersion except that it is sensitive to extreme values. That is why it is known as standard deviation. It is defined as the positive square root of the mean of the square of the deviations taken from mean ($\bar{x}$). It is denoted by **σ** . It is also known as mean error, mean square error or root mean square deviation from mean**.**

**Merit of standard deviation**

i. It is based on all items of the series.

ii. It is rigidly defined.

iii. It is least affected by fluctuation of sampling.

iv. It is suitable for further statistical treatments.



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v. It is suitable for comparing variability.

**Demerit of standard deviation**

ii. It can't be calculated for open end classes.
iii. It gives greater weight to the extreme values and less to those which are near the mean.
iv. It is difficult to compute and understand for non mathematical person.

**Calculation of standard deviation**

a. **For individual:** $\sigma = \sqrt{\frac{\sum(x-\bar{x})^2}{n}} = \sqrt{\frac{1}{n}\sum x^2 - \left(\frac{\sum x}{n}\right)^2}$

b. **For Discrete and continuous:** $\sigma = \sqrt{\frac{\sum f(x-\bar{x})^2}{N}} = \sqrt{\frac{1}{N}\sum fx^2 - \left(\frac{\sum fx}{N}\right)^2}$

where $N=\sum f$

E. **Varience:** Varience is just a square of standard deviation. i.e varience= $\sigma^2$.

**Varience** $= \frac{\sum f(x-\bar{x})^2}{N} = \frac{1}{N}\sum fx^2 - \left(\frac{\sum fx}{N}\right)^2$

**Co-efficient of varience (C.V ) :**

The relative measure of dispersion based on the standard deviation is defined as the ratio of the standard deviation and the mean is called coefficient of standard deviation.

Coefficient of SD $= \frac{\sigma}{\bar{x}}$

When coefficient of SD is expressed in percentage is called coefficient of variation, it is denoted by C.V. and given as:

C.V. $= \frac{\sigma}{\bar{x}} \times 100\%$

It is independent of unit of measurement. So two distributions can batter compared, with the help of C.V. for their variability. Less the C.V. more will be the uniformity, consistency, homogeneity, equitable etc. and more the C.V. less will be the uniformity, consistency, homogeneity etc.

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CHAPTER TW0

## CORRELATION AND REGRESSION ANALYSIS

**Correlation:** The degree of relationship between variable is called correlation. It may be simple, multiple or partial. Two variables are said to be in correlation when they are so related that the change in value of onevariable is accompained by change in value of other variable

**Types of correlation**

**a. Positive ,negative and No-correlation**

Two variables are said to be **positively correlated** if they tend to change together in the same direction, that is, if they tend to increase or decrease together. Such positive correlation is postulated by economic theory for the consumption and income. When the income increases the consumption increases, and conversely, when income decreases the consumption decreases

Two variables are said to be **negatively correlated** if they tend to change in the opposite direction: when X increases Y decreases, and vice versa. For example, the quantity of a commodity demanded and its price are negatively correlated. When price increases, demand for · commodity decreases and when price falls demand increases

Two variables are **uncorrelated** when they tend to change with no connection to each other. For example one should expect zero correlation between the height of the inhabitants of a country and the production of steel

**b. Linear and non linear correlation**

Correlation may be linear, when all points (X, Y) on a scatter diagram seem to cluster near a straight line, or nonlinear, when all points seem to lie near a curve

**c. Simple, partial and multiple correlation**

In simple correlation. we measure the correlation between two variables (of which one is dependent and the other is independent).

.

**Coefficient of correlation**

**a) Karl pearson's Co –relation coefficient**: The Karl Pearson's correlation coefficient measures the degree of linear association between two variables. This method is popularly known as Pearsonian correlation coefficient. It is unitless measure of correlation. It is defined as the ratio of covariance between two variables to the product of the standard deviation of the two variables. It is denoted by r, Let X and Y are two variables, then Karl Pearson's correlation coefficient between X and Y is given by:

$r = \frac{\text{covarience}(x,y)}{\sigma_x \times \sigma_y}$ where

$\text{co-varience}(x,y) = \frac{\sum(x-\bar{x})(y-\bar{y})}{n}$

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$\sigma_x = \sqrt{\frac{\sum(x-\bar{x})^2}{n}}$

$\sigma_x = \sqrt{\frac{\sum(y-\bar{y})^2}{n}}$

Where

$\bar{x}$ = The Arithmetic mean of values of variables X and

$\bar{y}$ = The Arithmetic mean of values of variables Y

and n = Number of pair of observations

$$r = \frac{n\sum xy - \sum x.\sum y}{\sqrt{n\sum(x)^2-(\sum x)^2} \times \sqrt{n\sum(y)^2-(\sum y)^2}}$$

**Properties of Karl Pearson's correlation coefficient**

Following are the important properties of Karl Pearson's correlation coefficient:

- Correlation coefficient lies between- 1 and + 1 i. e. - 1 < r < + 1
- The formula of Correlation coefficient is symmetrical i · e $r_{xy} = r_{yx}$.
- Correlation coefficient is geometric mean between two regression coefficient i.e. $r = \sqrt{b_{xy} b_{yx}}$



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- Correlation coefficient is a relative statistical measure and has no unit
- Two independent variables are uncorrelated but the converse may not be true 1.e. uncorrelated variables may not be independent.
- Correlation coefficient is independent of change of origin as well as of scale

**b) Spearman's correlation($r_s$):** The spearman's rank correlation is coefficient between x and y is given by $r_s = 1 - \frac{6\sum d^2}{n(n^2-1)}$ where the=difference of rank between x and y

**Merits of Spearman's Rank correlation Coefficient**

i. It is appropriate for qualitative data.
ii. It is easy to compute.
iii. It can be used when the data are highly skewed.

**Demerits of Spearman's Rank correlation Coefficient**

i. It is not suitable to compute in case of large number of observations.
ii. It is not appropriate for calculating correlation coefficient of a frequency distribution.
iii. It does not consider original values i.e. it involves ranks which is not actual measure.

**Coefficient of Determination:** The Square of the simple correlation coefficient is called the coefficient of determination. It is used for interpretation of the value of the calculated correlation coefficient.

If r is correlation coefficient then $r^2$ is coefficient of determination. If r =0.8, the value of $r^2$ = 0.64. this shows that 64% of the total variation in dependent variable has been explained by the independent variable and other 36% is due to other unknown factors. 1 - $r^2$ is called coefficient of non determination.

**Interpretation of calculated value of correlation coefficient**

The calculate value of Karl Pearson's correlation coefficient may be interpreted in the following manner.

- If r = 1 ; there is perfect positive correlation between the variables.
- If r = -1 ; there is perfect negative correlation between the variables.
- If r = 0;there is no correlation between the variables.
- If r lies between 0.700 to 0.999; there is high degree positive correlation.
- If r lies between - 0.700 to - 0.999; there is high degree negative correlation.
- If r lies between, 0.500 to 0.699;there is moderate positive correlation.
- If r lies between. - 0.500 to - 0.699; there is moderate negative correlation.
- If r lies between. 0.001 to 0.499; there is low degree positive correlation.
- If r lies between - 0.001 to - 0.499; there is low degree negative correlation.

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**Regression:** Regression is the average relationship among variables. It is used to predict the value of one variable on the basis of other variable or variables. Regresssion may be simple partial or multiple

**Lines of regression** The lines of regression are the lines, which gives the best estimated value of variables for the given value of independent variable's value. Let X and Y are two variables and there are two lines of regression then;

**1. Line of regression of y on x: (y=dependent and x=independent variable)**

Let,

$y = a+bx$..... (i)

be the line of regression of y on x; where a and b are constant to be determined.

By principle of least square method, the normal equation of (i) are

$\sum y = a\sum 1 + b\sum x$

$\sum y = na + b\sum x$.....(ii)

And

$\sum xy = a\sum x + b\sum x^2$ ..... (iii)

Solving equation (ii) and (iii), we get value of a and b

Let $a=\vec{a}$ ; $b=\vec{b}$

The fitted expression line of y on x is:


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$y=\vec{a}+\vec{b}x$

**Note:** For the line y=a+bx; b is called regression coefficient of y on x. It is denoted by $b_{yx}$.

**2. Line of regression of x on y: (x=dependent and y=independent variable)**

Let,

$x= a_1+b_1y$..... (i)

be the line of regression of x on y; where $a_1$ and $b_1$ are constant to be determined.

By principle of least square method, the normal equation of (i) are

$\sum y = a_1 \sum 1 + b_1 \sum y$

$\sum y = na_1 + b_1 \sum y$ ..... (ii)

And

$\sum xy = a_1 \sum y + b_1 \sum y^2$ ..... (iii)

Solving equation (ii) and (iii), we get value of a and b

Let $a_1=\vec{a_1}$

$b_1=\vec{b_1}$

The fitted expression line of x on y is:

$y=\vec{a_1}+\vec{b_1}x$

**Note:** For the line y=a+bx; b is called regression coefficient of x on y. It is denoted by $b_{xy}$.

**Properties of Regression Coefficients**

The following are the important properties of regression coefficients:

- Both regression coefficients must have same sign and the sign of correlation coefficient is same as the sign of regression coefficients.
- The product of the two regression coefficient must be less than or equal to 1 i.e. $b_{xy}$ X $b_{yx} \leq 1$
- The correlation coefficient is the geometric mean between two regression coefficient i.e. $r = \sqrt{b_{xy} + b_{yx}}$
- The two lines of regression intersect at the point (X ,Y ), where X and Y are the variables under study.
- Regression coefficient are independent of change of origin but not of scale
  i.e. $b_{xy} = b_{uv}$ if u = X- A and v = Y- B and
  $b_{xy} \neq b_{uv}$ if $u = \frac{X-A}{h}$ and $v = \frac{Y-B}{k}$ Where A and B are assumed mean, h and k are the common factor of X and Y series respectively.
- Arithmetic mean of regression coefficient is greater than the correlation coefficient.
- If $r = \pm 1$the lines of regression are either parallel or co-incident

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## CHAPTER THREE
## PROBABILITY

**Probability**: The probability can be defined as a measure of a likelihood that a particular event will occur; it is a numerical measure of chance factor which value lies between 0 and 1. If S is a sample space having 'n' points which arise out of all possible or exhaustive , equally likely and mutually exclusive outcomes of a random experiment and 'm' points in a S are Favourable to an event 'A', then the probability of the event A is given as

$$P(A) = \frac{\text{Favourable cases}}{\text{Total exhaustive number of cases}} = \frac{m}{n}$$

**Sample spaces and events:** The set of all possible outcomes of a random experiment is called **sample space.** The possible outcomes are called sample points. The sample space is usually denoted by the letter 'S'. For example:

- In a tosses of a single fair coin the sample points are head(H) and Tail (T) and sample space is S = {H, T}
- If two unbiased coins are tosses simultaneously , the sample space is S = { HH, HT, TH, TT}

Performing a random experiment is called a **trial** and the collections of possible outcomes are termed as **events**. For example; If a coin is tossed repeatedly, the result is not unique, we may get any of the two faces head or tail. Thus tossing of a coin is a random experiment or trial and getting of a head or tail is an event

### 3.1.1 TYPES OF EVENTS

**A. Simple or Compound Events:** An event having only one sample point is called **simple or elementary event**. Otherwise it is known as **compound or composite event**. Thus in tossing of a single die the event of getting '5' is a simple event but the event of getting an odd number is a compound event.

**B. Mutually Exclusive Events :** Events are said to be **mutually exclusive or disjoint** if the occurrence of one precludes the occurrence of other i.e. one and only one of them can occur in a single trial. For example in tossing of a fa ir coin, the events head and tail are mutually exclusive events since if the coin fall with head upside tail can't turn up and vice-versa. If two or more events can occur together at a time they are not mutually exclusive. For example, if we draw either a king or a diamond from a pack of 52 cards then event king and diamond can occur together because we could draw the king of diamond, the king and diamond are not mutually exclusive.

**C. Exhaustive Events** :The total number of possible outcomes of a random experiment is known as the **exhaustive number of event**. For example, in tossing of a fair coin exhaustive cases are head and tail. In tosses of two fair coin at a time th exhaustive cases are (HH, HT, TH and TT}

**D. Favourable Events**: The number of cases Favourable to an event in a trial is the number of outcomes which entail the happening of the event. For example in drawing a card from a pack of cards the number of cases Favourable to drawing of a king is 4, for drawing of diamond is13 and for drawing a black card are 26.

**E. Equally Likely Events :**Outcomes of a trial are said to be equally likely if none of them is expected to occur in preference to other i.e. happening of all event has equal and independent chance. For example in throwing an unbiased die, all the six faces are equally likely to come; in a random toss of a fair coin head and tail are equally likely.

**F. Independent and Dependent Events:** Two or more events are said to be **independent** if the occurrence or non-occurrence of one event doesn't affect the occurrence or non-occurrence of the other events. Otherwise said to be **dependence**. For example, if we toss 2 coins together, let $E_1$ be the event of occurrence of head or tail on first coin and $E_2$ be the event of occurrence of head or tail on second coin , so $E_1$ and $E_2$ are two independent events. If we draw a card from a pack of well-shuffled cards and replace it before drawing the


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second card, the result of the second draw is independent of the first draw, but, however, if the first card drawn is not replaced then the second draw is dependent on the first draw.

**Axioms, interpretations and properties of probability**

If S is a sample space having 'n' points which arise out of all possible or exhaustive , equally likely and mutually exclusive outcomes of a random experiment and 'm' points in a S are Favourable to an event 'A', then the probability of the event A is given as

$$P(A) = \frac{m}{n}$$

A. In probability n and m are positive integer , thus , we have $0 \leq m \leq n$. $0 \leq m \leq 1 \therefore 0 \leq p(A) \leq 1$, hence the probability of an event A is a number which is always lies between 0 and 1.
B. If P(A) = 0, then A is called an impossible or null event and If P(A) = 1, then A is called a certain or sure event.
C. Probability is a pure number, it has no unit.
D. Let A be event of non- occurrence of an event A then,
$$P(\bar{A}) = \frac{(\text{Number of favourable cases for non-occurence of A})}{\text{Total cases}} = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(A)$$
   a. $P(A) + P(\bar{A}) = 1$, ie total probability is always equal to unity.
E. We can compute the probability of an event A, i. e. P (A) by logical reasoning without conducting any experiment, since, the probability, P (A) can be computed prior to obtaining any experimental data, and it is also termed as a priori or mathematical probability.

**Conditional probability**: Many times the information is available that a event has occurred and one is required to find out the probability of occurrence of another event B, utilizing the information about A , such a probability is known as conditional probability and is denoted by P(B/A) i.e. the probability of the event B given A For example, suppose we know that a accident will happen (A) then one is interested to know the probability of death of patient (B) i.e. we want to calculate P(B/A) ,

The formula is $P(B/A) = \frac{P(A \cap B)}{P(A)}$

If A and B are independent, then $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A).P(B)}{P(A)} = P(B)$

**Theorems on probability(Addition, Multiplication and bayes and their applications)**

The theorems of probability are as follow:

**Additive theorem of probability:** There are two cases arise for additional theorems of probability.They are:

**Additive theorem of probability for not mutually exclusive events:** T he probability of occurrenceof at least one of the two events A and B is given by:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

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**Proof**

Let us suppose that a random experiment results in a sample space S with N sample points, then by definition

$$P(A \cup B) = \frac{n(A \cup B)}{n(S)} = \frac{n(A \cup B)}{N}$$

Where $n(A \cup B)$ is the number of occurrences favorable to the event $(A \cup B)$ . From the adjoining diagram we get.

$$P(A \cup B) = \frac{n(A) - n(A \cap B) + n(B)}{N} = \frac{n(A)}{N} - \frac{n(A \cap B))}{N} + \frac{n(B)}{N} = P(A) + P(B) - P(A \cap B)$$

**Additive theorem of probability for mutually exclusive event:** If the events are mutually disjoint i.e. $(A \cap B) = \Phi$ then $P(A \cap B) = 0$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B)$

Hence probability of happening of any one of the two mutually exclusive events is equal to the sum of their individual events.

**Multiplication theorem of probability:** There are two cases aries for multiplication theorems of probability. They are:



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**Multiplication theorem of probability for dependent events**: The probability of simultaneoushappening of two events A and B is given by:

$P(A \cap B) = P(A)\, P(B/A);\ P(A) \neq 0$

$P(A \cap B) = P(B)\, P(NB);\ P(B) \neq 0,$

where P(B/A) is the conditional probability of happening of B· under the condition that A has already happened.

**Proof**

Let A and B be the events associated with the sample space S of random experiment with exhaustive number of outcomes N ie n(S) = N, then by defination

$\mathbf{P(A \cap B) = \frac{n(A \cap B)}{N}}$ .For the conditional event A/B , the Favourable outcomes must be out of the sample point B. In other words for the event A/B, the sample space is B and then

$P(A/B) = \frac{n(A \cap B)}{n(B)}$.

Similarly $P(B/A) = \frac{n(A \cap B)}{n(A)}$

$\therefore \mathbf{P(A \cap B) = \frac{n(A \cap B)}{N} = \frac{n(A)}{N} x \frac{n(A \cap B)}{n(A)}} = P(A) \times P(B/A)$

$\therefore \mathbf{P(A \cap B) = \frac{n(A \cap B)}{N} = \frac{n(B)}{N} x \frac{n(A \cap B)}{n(B)}} = P(A) \times P(A/B)$

**Multiplication theorem of probability for independent events:** If A and B are independent, so thatthe probability of occurrence of or non-occurrence of A is not affected by the occurrence or non- occurrence of B, then we have

$P(A/B) = P(A)$ and $P(B/A) = P(B)$

$\therefore P(A \cap B) = P(A)\, P(B)$

**Bayes theorem and their application**

**Statements:** If $E_1, E_2, E_3 \ldots\ldots E_n$ are and mutually and exclusive events of samples space "s" such that $\bigcup_{i=1}^{n} E_i = s$ and $P(E_i)>0$ for i=1,2,3…… and and if A is any orbitrary events which is associated with $E_1$'s such that P(A)>0. Then the probability that A has occurred and given by the event $E_i$ is given as P(E/A) i.e.

$P(E/A) = \frac{P(A/E_i)P(E_i)}{\sum_{i=1}^{n} P(A/E_i)P(E_i)}$

**Proof:** The events $E_1, E_2 \ldots\ldots E_n$ are mutually exclusive events and A is any arbitrary event which is associated with $E_1$ ' and $P(A)>0$ & $P(E_i)> 0$. Then from the conditional probability of two events A and $E_i$

$P(A \cap E_i) = P(A) \times P(E/A) = P(E_i) \times P(A/E_i)$

or $P(E_i/A) = \frac{P(E_i) x P(A/E_1)}{P(A)}$ ..... (i)

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The event A is define as

$A=(A \cap E_1) \cup (A \cap E_2) \cup \ldots\ldots\ldots \cup (A \cap E_n)$,

since each $E_i$ are mutually exclusive and thus $(A \cap E_1), (A \cap E_2), \ldots\ldots\ldots, (A \cap E_n)$ are also mutually exclusive.

$\therefore P(A) = P(E_1) \times P(A/E_1) + P(E_2) \times P(A/E_2) + P(E_3) \times P(A/E_3) + \ldots\ldots\ldots P(E_n) \times P(A/E_n)$

$P(A) = \sum_{i=1}^{n} P(E_i) x P(A/E_1)$ .....(ii)

From Equation (i) and (ii)

$P(E_i/A) = \frac{P(E_i) x P(A/E_1)}{\sum_{i=1}^{n} P(E_i) x P(A/E_1)}$

For three exclusive and exhaustive events,

$P(E_1/A) = \frac{P(E_1) x P(A/E_1)}{P(E_1) x P(A/E_1) + P(E_2) x P(A/E_2) + P(E_3) x P(A/E_3)}$

$P(E_2/A) = \frac{P(E_2) x P(A/E_2)}{P(E_1) x P(A/E_1) + P(E_2) x P(A/E_2) + P(E_3) x P(A/E_3)}$

$P(E_3/A) = \frac{P(E_3) x P(A/E_3)}{P(E_1) x P(A/E_1) + P(E_2) x P(A/E_2) + P(E_3) x P(A/E_3)}$


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CHAPTER FOUR

## RANDOM VARIABLES AND MATHEMMATICAL EXPECTATION

### Random variables

A rule that assigns a real number to each outcome of a random experiment than the set of this real number is called a random variable (r v).Thus random variable associates numerical values with each elementary outcomes of an experiment. A random variable is usually denoted by any of the Capital Latin letters X, Y, Z, U, V ... etc and particular values which the random variable takes are denoted by the corresponding small letters **.** If we flip a coin and denote the head by 1 and tail by 0 then the random variable takes place the value 1 and 0. In symbol X={x:x=(1,0)є S}

### Types of Random Variables

There are two types of random variable namely.

A. **Discrete random variable**: A random variable takes a finite number of values or an infinite number of values that can be arranged in a sequence that can be counted with positive integers 1,2.....And so forth is known as Discrete random variable. Further if there is a natural gap between two values of the variable then the variable is discrete random variable. For example:

a) If we toss a coin, the variable can take only two values 0 and 1 assigned to tail and head respectively.
b) The number of student in a class etc
c) The number of people in queue in a Cinema hall for ticketing,
d) The number of children in a family,
e) Number of run made by a Cricketer in a match etc

B. **Continuous random variable**: A random variable X, which can take all possible values in Real line R or between certain limits is said to be a continuous random variable. For a continuous random variable the probability at a point is zero. For example:

a) The weight of middle aged people in Kathmandu lying between 40 kg and 150 kg is a continuous random variable. Symbolically $X = \{x: 40 \leq x \leq 150\}$,
b) Measurement of weight, height, temperature, electric voltage, rainfall,
c) Hardness of steel,
d) Distance covered by a vehicle etc .

### Properties of Random variables

The important properties of random variable are as follows

a) If X is a random variable and a and bare any two constants, then aX+b is also a random variable
b) If X is a random variable, then $X^2$ is also a random variable
c) If X is a random variable then 1/ X is also a random variable
d) If X and Y are random variables defined on the same sample space S then, X+Y, X-Y, aX, bY or aX+bY are also random variables with a and b are non negative constants
e) If $X_1, X_2 \ldots\ldots X_n$ are n independent random variables then $U_n = \min(X_1. X_2. \ldots\ldots X_n)$ and $V_n = \max(X_1. X_2 \ldots\ldots X_n)$ are also random variables

### Probability mass function(P.M.F)

If X is a one-dimensional discrete random variable taking at most a countable numbers $x_1, x_2 \ldots\ldots x_n$ with each value of the variable X , we associate a number

$P(x_i) = P(X = x_i)$ ; i = 1, 2,.... , n.

This is known as the probability of variate values $x_i$ and satisfying the following two conditions

i . Each $P(X_i) = P(X = x_i) \geq 0$ i.e. $P(x_i)$ are all non-negative and

ii. $\sum P(x_i) = 1$ i.e. the total probability is unity.

Then $P(x_i) = P(X = x_i)$ is called probability mass function of random variable X. The set of ordered pairs $\{x;, P(x_i)\}$ i = 1, 2, ....,n is called the probability distribution of the random variable X

### Probability density function

If X is continuous random variable and f(x) is a continuous function of X. Then f(x)dx gives the probability of the event that random variable X lies in the interval $(x - \frac{dx}{2}$ and $x + \frac{dx}{2})$

i.e $f(x)dx = P(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2})$



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Then f(x) is called probability density function (p.d.f.). It is also known as **frequency function** because it also gives the proportion of units lying in the interval $x - \frac{dx}{2}$ and $x + \frac{dx}{2}$. If X has the range [a, b]. i.e. a≤X≤ b then the probability density function of random variable X must satisfy the following two Properties

f(x) ≥0 for every x ε [a, b]

$\sum_a^b f(x)dx = 1$,

it implies that the total area under the frequency curve is always unity.

**Cumulative distribution function:** A function $F_x(x)$ of a random variable X for a real value x giving the probability of an event (X≤x) is called a cumulative distribution function(c.d.f). it is simply known as distribution function.

**Properties of Distribution of function**

Some importance properties of distribution

**a.** If a and b are two constant values such that a < b and F is the distribution function then
P(a <x≤b) = F(b)- F(a)
**Proof** :The events 'a < X ≤b' and 'X≤a' are disjoint and their union is the event 'X≤b'. Hence by addition theorem of probability
P(a<X ≤b)+ P(X≤ a) = P(X ≤b)
P(a<X ≤b) = P(X≤ b) - P(X: ≤ a)
P(a <X ≤b) = F(b)- F(a)

**b.** If F(x) is the distribution function of a mono-variate X, then 0≤*F(x)* ≤1

**c.** If F(x) is the distribution function of mono-variate X , then $F(-\infty) = \lim_{n\to-\infty} f(x) =$ 0 *and* $F(\infty) = \lim_{n\to\infty} f(x) = 1$

**Mathematical expectation**

**Physical meaning of mathematical expectation:**Let X be a one dimentional descrete Random variables which can assume any one of the values $x_1, x_2 .. x_n$, with respective probabilities P ($x_1$) , P ($x_2$) ... P ($x_n$). Then the mathematical expectation of r v X usually called the expected value of X and denoted by E(X) and it is defined as

$E(X) = x_1P(x_1) + x_2P(x_2) + ..... + x_nP(x_n) = \sum x\, P(x)$

Where $\sum_{i=1}^n p(x) = 1$ i.e. total probability is unity

Similarly, if X is continuous random variable with probability density function f(x) then mathematical expectation of r. v X is defined as

$E(X) = \int x f(x)\, dx$; where f(x) ≥0 for every xϵ[a, b] and $\int_a^b f(x)dx = 1$

**Physical meaning of Mathematical Expectation**

Let us consider the following frequency distribution of the discrete random variable X as follwos

| X | $x_1$ | $x_2$ | … | $x_n$ |
|---|---|---|---|---|
| F | $f_1$ | $f_2$ | … | $f_n$ |

Then the arithmetic mean of the discrete random variable is given by

$$\bar{x} = \frac{\sum fx}{N} = \frac{f_1x_1+f_2x_2+\cdots+f_nx_n}{N}$$

$$\bar{x} = \frac{f_1}{N}x_1 + \frac{f_2}{N}x_2 + \cdots + \frac{f_n}{N}x_n$$

We observed that, out of total of N cases f; cases are favourable to each of x events.

Therefore $P(X = x) = \frac{f_1}{N} = P(x_i)$, i = 1, 2 ..... n.

Which implies that, $\frac{f1}{N} = P(x_i)$, $\frac{f_2}{N} = P(X_2)$ ..... ..... . $\frac{f_n}{N} = P(x_n)$

Therefore, $x = x_1P(x_1) + x_2P(x_2) + ..... + x_nP(x_n) = \Sigma xP(x) = E(X)$.


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Hence, mathematical expectation of a random variable is nothing but its simply arithmetic mean of random variable.

**Properties of Mathematical Expectation**

a. If 'k' is a constant then E(k) = k
b. The mean of random variable Y defined as Y = a x + b with mean of random variable X as μ is
   = a E(X) + b = aμ + b
   'a' and 'b' are two constants then E (aX +b) = a E(X) + b
   E (aX+ b)= E(aX) + E (b)
   =a E(X) + b
   =a E(X) + b
c. The mathematical expectation of the sum of random variables is equal, to the sum of their expectation, provided all the expiation exists. Symbolically, Let X, Y Z ... T are random variables then
   E ( X + Y + Z + ... + T) = E ( X ) + E ( Y) + E ( Z ) + ... + E ( T)

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## CHAPTER FIVE
## DISCRETE PROBABILITY DISTRIBUTION

### Binomial and possion probability distribution

### Binomial Probability distribution:

❖ **Derivation of probability mass function of Binomial distribution**

The probability of x successes and consequently (n-x) failures in 'n' independent trials, in a specified order say

FFSFSSSF .....SFS ,

where S represent success and F represents failure, is given by the compound probability theorem as

P (F F S F S S S F.... S F S)

= P (F) P(F) P(S) P(F) ....P(S) P(F) P(S)

= q q p q.... p q p. [Since trial are independent]

= ppp… … ... p qqq.... qqq

( x successes) (n-x failures)

$= p^x q^{n-x}$

But x success in 'n' trials can occur in ${}^nC_x$ ways and the probability for each of these ways is same (say) = $p^x q^{n-x}$

Hence the probability of x success in n trials in any order is given by the addition theorem of probability by the expression ${}^nC_x\, p^x q^{n-x}$.

Here ${}^nC_x\, p^x q^{n-x}$ is the $(x+1)^{th}$ terms in the binomial expansion of $(p+q)^n$.

❖ **Characteristics of Binomial Probability distribution**

- The two independent constant and and p in the binomial distribution are known as the parameters of the binomial distribution , sometimes 'n' is also known as the degree of the binomial distribution.
- Binomial distribution is a discrete distribution as X can take only the integral values viz 0, 1, 2, 3 ... ..... ..... . n.
- Any random variable which follows binomial distribution is known as binomial variate.
- We shall use the notation X≈ B(n, p) to denote that the random variable X follows binomial distribution with parameter and and p.
- The P(x) defined in binomial distribution is probability mass function bacause

$$\sum_{0}^{n} p(x) = \sum_{0}^{n} p^x q^{n-x} = 1$$

- For the binomial distribution, variance is less than mean.
  Since q is the probability of failure, for binomial distribution we always have
  0 < q < 1, multiply by np on all sides we get
  0<npq<np
  0 < variance < mean
  Thus, for binomial distribution mean is greater than that of variance.
- If $p = q = \frac{1}{2}$ the binomial distribution is symmetrical .
- If $p < \frac{1}{2}$, t e binomial distribution is positive skewed.
- If $p > \frac{1}{2}$, the binomial distributicn is negative skewed.
- Binomial distribution is unimodal if 'np' is a whole number and the mean and mode are equal, each being 'np'

❖ **Mean and variance of binomial distribution**

Let X follows binomial distribution with parameters and n and p then $P(X = x)\begin{cases} {}^nC_x p^x q^{n-x} \text{ for } x=0,1,2.....n \\ 0 \text{ otherwise} \end{cases}$

$\therefore$mean =E(x)= $\sum x\, p(x) = \sum_{x=0}^{n} x\, {}^nC_x p^x q^{n-x} =$

$= \sum_{x=0}^{n} \frac{xn!}{(n-x)!x!} p^x q^{n-x}$


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$$= \sum_{x=0}^{n} x \frac{n(n-1)!}{(n-x)!x(x-1)!} p^x q^{n-x}$$
$$= np \sum_{x=1}^{n} {}^{n-2}C_{x-2} p^{x-1} q^{n-x}$$
$$= np(q+p)^{n-1}$$
$$= np$$

Thus mean of binomial distribution is np.

Similarly, variance of X= var(x)=$E(x^2) - [E(x)]^2$

then, $E(x^2)=\sum x^2 P(x) = \sum_{x=0}^{n} \{x(x-1) + x\} p(x) = \sum_{x=0}^{n} x(x-1) {}^{n}C_{x} p^x q^{n-x} + p(x)$

$$= \sum_{x=0}^{n} \frac{n(n-1)(n-2)! p^x q^{n-x}}{(n-x)!(x-2)!} + np$$
$$= n(n-1)P^2 \sum_{x=2}^{n} {}^{n-2}C_{x-2} p^{x-2} q^{n-x} + np$$
$$= n(n-1)P^2 (p+q)^{n-2} + np = n(n-1)p^2 + np$$

∴ variance of X= var(x)=$E(x^2) - [E(x)]^2$

$= n(n-1)p^2 + np - \{np\}^2$

$= np(1-q) = npq$

Thus Varience of binomial distribution is npq.

**Poisson Probability Distribution:**

When the number of trials and is very large and probability of success is very small then the resulting distribution is known as poisson distribution. It deals with the evaluation of probabilities of rae events such as

- Number of car accidents on road
- Number of earthquakes in a year
- Number of printing mistakes per page
- Number of blind children born in a city
- The number of twin births in a city hospitals etc.

❖ **Conditions of Poisson distribution**

Poisson distribution is a limiting case of binomial distribution under the following conditions.

- The discrete random variable X is the number of occurrence of ' and event in an interval of time.
- The occurrences of the events must be random.
- The occurrences of the events must be rare and independent of each other.
- The occurrence must be uniformly distributed over the interval being used.
- The number of trials 'n' is indefinitely large i.e. $n \to \infty$
- The constant probability of success 'p' for each trial is indefinitely small i.e. $p \to 0$
- The product of n and p i.e. $n\,p = \lambda$ (say) is finite so that $P = \frac{\lambda}{n}$ and $Q = 1 - \frac{\lambda}{n}$ where λ is read as lambda.

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❖ **Definition of Poison Distribution**

A discrete random variable X is said to follows a poison distribution with single parameters A if it assumes only non-negative values and it's probability mass function is given by

$$P(X = x) \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & \text{for } x = 0,1,2 \ldots \ldots n \\ 0 & \text{otherwise} \end{cases}$$

❖ **Properties/characteristics of Poisson distribution**

Following are the important properties of Poisson distribution.

- The Poisson distribution depends upon the value of A , the average number of success per unit, 1·. is called the parameter of the Poisson distribution.
- Like binomial distribution Poisson distribution is a discrete probability distribution, it is concerned with occurrence that can take ·integral value 0,1 , 2 ... .
- Any random variable which follows Poisson distribution is known as Poisson variate.
- We shall use the notation X- P(λ) to denote that random variable X follows Poisson distribution with parameter λ.



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- The probability of success P(X=x) = P(x) defined in the Poisson distribution is pmf because

$$\sum_{x=0}^{\infty} \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\sum_{x=0}^{\infty}\frac{\lambda^x}{x!} = e^{-\lambda}e^{\lambda} = 1$$

- Poisson distribution is independent of trial n.
- Poisson variates are independent to each others.
- The Poisson distribution is a positively skewed distribution. As λ increase, the graph of Poisson distribution would get closer to a symmetrical continuous curve.
- Poisson distribution is unimodal if λ is not an integer and unique modal value is integral part of A and it is bimodal if λ is a integral and models value being λ and (λ -1 )

## ❖ Means and variance of poisson distribution

The probability mass function of Poisson distribution is

$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$ x=0,1,2,3……∞

mean = E(x)= $\sum x\,p(x)$

$= \sum_{x=0}^{n} \frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\lambda\sum_{x=1}^{n}\frac{\lambda^{x-1}}{(x-1)!}$

$= e^{-\lambda}\lambda e^{\lambda} = \lambda$

Thus mean of poissons distribution is λ

Similarly,

variance of X= var(x)=$E(x^2) - [E(x)]^2$

then, $E(x^2)=\sum^{\infty} x^2P(x) = \sum^{\infty}\{x(x-1)+x\}p(x)$

$= \sum_{x=0}^{\infty} x(x-1)\frac{e^{-\lambda}\lambda^x}{x(x-1)(x-2)!} + \sum_{x=0}^{\infty} xp(x)$

$= e^{-\lambda}\lambda^2\sum_{x=0}^{\infty}\frac{\lambda^{x-2}}{(x-2)!} + \lambda = e^{-\lambda}\lambda^2 e^{\lambda} + \lambda$

$= \lambda^2 + \lambda$

∴ variance of X= var(x)=$E(x^2) - [E(x)]^2$

$= \lambda^2 + \lambda - \lambda^2 = \lambda$

∴ Varience of binomial distribution is λ.

Hence for poissons distribution the mean and variance is equal to λ.

### Fitting of Binomial and poisson Distribution

### Fitting of Binomial Distribution

Let X follows binomial distribution with parameter n and p then

$P(X=x)={}^nC_x p^x q^{n-x}$

$P(X=x+1)={}^nC_{x+1} p^{x+1} q^{n-x-1}$

hence,

$\frac{P(X=x+1)}{P(X=x)} = \frac{{}^nC_{x+1}p^{x+1}q^{n-x}}{{}^nC_x p^x q^{n-x}}$

$\therefore P(X=x+1)=\frac{(n-x)p}{(x+1)q}\,P(X=x)$

This relation is used to calculate the probabilities of binomial distribution when probabilities of binomial distribution when probability of X=0 is known.

If a trail is repeated for N times then theoritical frequency or expected frequency is given by

$F(x)=N\times P(x) = N\times {}^nC_x\, p^x q^{n-x}$ x=0,1,2,3,….n

This is known as **fitting of binomial distribution**.

### Fitting of poisson Distribution

For poission distribution with parameter λ

$P(X=x)= \frac{e^{-\lambda}\lambda^x}{x!}$ , x=0,1,2 ……∞


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$P(X=x+1)=\frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}$ , x=0,1,2 ……∞

$$\frac{P(X=x+1)}{P(X=x)}=\frac{\frac{e^{-\lambda}\lambda^{x+1}}{(x+1)!}}{\frac{e^{-\lambda}\lambda^{x}}{x!}}=\frac{\lambda}{(x+1)}$$

$\therefore P(X=x+1)=\frac{\lambda}{(x+1)}P(X=x)$ which is required reccurrence formula for poission distribution, used to calculate the probability of poissions distribution when probability of X=0 is known.

If an experiment, satisfying the requirement of poission distribution is repeated N – times then an expected frequency or theoritical frequency of getting x – successes is given by

$F(x)=N\,P(X=x)=N\frac{e^{-\lambda}\lambda^{x}}{x!}$, x=0,1,2,3…∞

This is known as fitting of poission distribution

**The hyper-geometric and negative binomial distribution**

**The hyper-geometric distribution**

A discrete random variable X is said to follow the hyper geometric distribution with parameters N, M and n if it assumes only non-negative value and its probability mass function is given by

$$P(X=x)=\begin{cases}\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} & x=0,1,2,3\ldots\ldots \min(n,M):\ M\leq N:\ n\leq \text{And}\\ 0 \text{ otherwise}\end{cases}$$

Where N is a positive integer, M is a positive integer not exceeding And and n is a positive integer that is at-most N.

Under the following condition we use hyper-geometric distribution.

- The population or universe where the sample drawn consist of N finite individual, objects or elements.
- Each individual can be characterized as a success or failure and then are M successes in the whole population.
- A random sample of n individuals is selected without replacement In such a way that each subset of size n is equally likely to be chosen.

❖ **Properties of Hyper-geometric Distribution**

The following are the properties of Hyper-geometric distribution

- The three independent constants, N, M and n in the distribution art, known as the parameters of hyper geometric distribution
- Hyper geometric distribution is a discrete distribution as X can take only the integer values viz, 0, 1, 2 .... .. min(n. M).
- Any random variable which follows hyper geometric distribution is known as hyper geometric variate.We shall use the notation X-HG (k; N, M, n) to denote that the random variable X follows hyper geometric distribution with three parameter, And M and n.
- The P(X = x) defined in the hyper geometric distribution is in pmf because

$$=\sum_{x=0}^{n}\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}=1$$

- Mean and varience of hyper geometric distribution are $\frac{nM}{N}$ and .....respectively

**Negative Binomial Distribution**

Negative binomial distribution is the converse of binomial distribution. In binomial distribution the number of trials are fixed but in negative binomial distribution the number of success is fixed and the number of trial is random variables.

The probability of x-failure preceding the $r^{th}$ success is given by

$P(X=x)=\binom{-r}{x}(p)^{r}(-q)^{x}$ , x=0,1,2,3….

which is the $(x+1)^{th}$ term in the expansion of $p^{r}(1-p)^{r}$.

Negative binomial distribution can be used under the following conditions.



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- The experiment consists of sequence of independent trials.
- Each trial has only two possible outcomes; a success and a failure.
- The probability of success is constant from trial to trial.
- The experiment continues until a total of 'r' success have been observed with last trial success
- We count the number of trials before the success occurs.

### ❖ Properties of Negative Binomial Distribution

The properties of Negative Binomial Distribution are as follows:

- The negative binomial distribution depends upon 'r' and 'p' then 'r' and 'p' are called the parameter of the negative binomial distribution.
- Negative binomial distribution is a discrete probability distribution. It is concerned with occurrence that can take integral values as 0, 1, 2 ...
- Any random variable which follows negative binomial distribution is known as negative. binomial variate.
- We shall use the notation X-NBD(r, p) to denote the random variable X follows negative binomial distribution with parameters 'r' and 'p'.
- The P(X=x) defined in NBD is pmf because

$$\sum_X p(x) = \sum_x {}^{-r}C_x p^r(-q)^x = p^r \sum {}^{-r}C_x(-q)^x = p^r(1-q)^r = 1$$

- Mean and variance of negative binomial distribution are $\frac{rq}{p}$ and $\frac{rq}{p^2}$ respectively
- In Negative Binomial Distribution the number of success is fixed and number of trials is a random variable.
- Variance of Negative Binomial Distribution is greater than its mean i.e
  $\frac{rq}{p^2} > \frac{rq}{p}$ => $p < 1$, which is true.

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## CHAPTER SIX

## CONTINUOUS PROBABILITY DISTRIBUTION

**Normal, standard normal, the gamma, chi square and t-distribution**

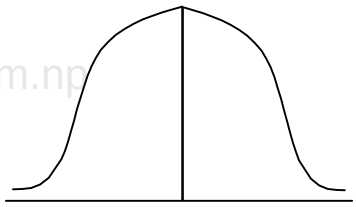
**Normal distribution**

A continuous random variable X is said to have a normal distribution with parameters μ called mean and $\sigma^2$ called variance, if its pdf is given by the probability law as

$$f(x)=f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(X-\mu)^2}{2\sigma^2}};\ -\infty< x<\infty;\ -\infty<\mu\ \&\ \sigma>0$$

**Characteristics of normal distribution and normal probability curves**

- The normal probability curve with mean μ & variance $\sigma^2$ is given by the equation f(x) $=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(X-\mu)^2}{2\sigma^2}}$ and has the following properties
- The graph of probability f(x) of normal distribution is famous 'bell – shaped' curve. The top of the bell is directly above the mean μ and tails of curve of a normal distribution curve extended indefin itely on both
- sides of X = μ and never touch the x- axis. For large value of σ, the curve tends to flatten out and for small value of $\sigma$ it has a sharp peak.
- Mean, median and mode of the distribution are coinciding.
- The two independent constants p and $a$ in the distribution are known as the parameters of the distribution.
- Normal distribution is a continuous distribution as x can takes any value in the real line R i.e. $-\infty < x < \infty$
- Any random variable which follows normal distribution is known as normal variate.
- We shall use the notation x - N (μ , $\sigma^2$) to denote that the random variable X follows normal distribution with mean p and variance $\sigma^2$
- The probability function f(x) defined in the distribution is pdf because $\int_{-\infty}^{\infty} f(x) = \text{i.e} = \int_{-\infty}^{\infty}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(X-\mu)^2}{2\sigma^2}} = 1$
- The normal curve is symmetrical about the line X = μ , thus coefficient of skewness $\beta_1 = 0$ and $\lambda_1 = 0$. the normal distribution is masokurtic thus $\beta_2 = = 3$ and $\lambda_2 = 0$.
- Since f(x) being the probability, can never be negative, no portion of the curve lies below the x - axis.
- Linear combination of independent normal variates is also a normal variate.
- X-axis is an asymptote to the normal curve.
- Mean deviation about mean=$\sqrt{\frac{2}{\pi}}\sigma = \frac{4}{5}\sigma$(approx.)
- Quartiles are $Q_1 = \mu - 0.6745\,\sigma$, $Q_3 = \mu + 0.6745$
- Quartile deviation Q.D. = $(\frac{Q_2-Q_1}{2})$(approx)
- Therefore QD: MD: SD = $\frac{2}{3}\sigma : \frac{4}{5}\sigma : \sigma$
- Which implies QD:MD:SD = 10: 12: 15



X=μ

Fig: Normal probability curve

**IMPORTANCE OF NORMAL DISTRIBUTION**

Normal distribution plays a very important role in statistical theory because of the following reasons.

1. For large values of n, calculations of probability for discrete probability distribution (e.g. Binomial distribution, Poisson distribution, hyper geometric distribution etc,) become quite tedious and time consuming. In such cases, normal approximation can be used with great ease and convenience.
2. For large value of n (i.e. n →∞) , almost all of the exact sampling distribution e.g.- t-distribution , f-distribution and the chi-square distribution conform to normal distribution.
3. The whole theory of exact sample (small sample) tests namely t. f ,$x^2$ tests etc is based on the assumption that the parent population from which the sample are drawn follows normal distribution.
4. For large value of n (i.e. n →∞), the central limit theorem follows normal distribution.



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5. It is extensively use in large sampling theory to find the estimates of parameters from statistic andconfidence limits etc
6. Many of the distributions which are not normal can be made normal by simple transformation.
7. It finds many applications in statistical quality control and industrial experiments for setting control limits.

**Standard normal distribution**

If X is a normal variate with parameter μ and $\sigma^2$ then random variable Z defined by

$Z = \frac{x-\mu}{\sigma}$ is a Standard normal variate with mean zero (0) and Cr variance unity (1) and the pdf of Standard normal distribution is given by

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, -\infty < Z < \infty$$

**The gamma distribution**

A continuous random variable X is said to have a gamma distribution with single parameter λ, if it assumes only non- negative values and it probability density function is defined as $f(x) = f(x,\lambda) =$

$$\begin{cases} \frac{e^{-x} x^{\lambda-1}}{(\lambda)} & \lambda > 0, 0 \le x < \infty \\ 0 \ otherwise \end{cases}$$

**Properties of gamma distribution**

Following are the important properties of gamma distribution.

- The Game distribution depends upon the two constants α and.λ are called the parameter of the Gamma distribution.
- Like normal distribution gamma distribution is also a continuous probability distribution
- Any random variable which follows gamma distribution is known as gamma variate.
- We shall use the notation X = G(α,λ) to denote that random variable X follows Gamma distribution with parameters α and λ. and also X = G(λ)for single parameter A.
- The f{x, λ.) defined in gamma distribution is pdf because

$$\int_0^\infty f(x,\lambda) = 1$$

- As λ→∞ gamma distribution tends to normal distribution.
- If $x_1, x_2 \ldots\ldots\ldots\ldots X_n$ are identically independently distributed (iid) gamma variate with parameter $\alpha, \lambda_i$; i= 1, 2 .... ..... n then $\sum x_i \approx G(a, \sum \lambda_1)$

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**Chi square distribution**

The square of a standard normal variate is known as a chi-square variate (pronounced as ki - sky without s) with **1** degree of freedom (d.f.). Thus if Z is SNV with mean 0 and variance 1 than Z is known as the chisquare variate.

In general if $x_1, x_2 \ldots\ldots\ldots\ldots x_n$ are n independent normal ~arlnt •, wltll mean J..l; and variances. cr; (j = 1' 2 .. ... n), then $Z_1, Z_2$ ----------- $\cdot Z_n$ are independent SNV& $x^2 = z_1^2 + z_2^2 + z_3^2 + \cdots \ldots\ldots\ldots + z_n^2$ is chi-square variate with n. df and its pdf is defined as

$$f(x^2) = \frac{1}{2^{\frac{1}{2}} \Gamma_{\frac{n}{2}}} e^{-\frac{x}{2}} x^{\frac{n}{2}-1} \quad ; 0 \le x < \infty$$

**Properties of Chi-Square Distribution**

- A random variable X which follows chi-square distribution is also known as chi-square variate.
- A chi-square variate X is continuous random variable which assumes non-negative variables only i.e. o ~ x < oo.


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- We shall use the notation X "'l(n) to denote that random variable X follows chi- square distribution with n- degrees of freedom.
- Mean and variance of chi square distribution are nand 2n respectively.
- Mode of the chi-square distribution curve is lies at the point $x^2$=and $-2$ for n > 2
- Measure of Skewness $\frac{8}{n}$and Kurtosis is$\frac{12}{n}$ Therefore, Chi-square curve is positive Skewness & Leptokurtlc
- The chi-square distribution curve with 1 and 2 degree of freedom is hyperbolic in shape where as for n > 2, the curve is un-symmetrical shaped.
- If $x_1$ ,$x_2$, ... ..... $x_k$ are k iid chi square variate with $n_i$ d.f then ;$\sum x$ in also a chi-square variate with $\sum n_2$ d.f. this is known as additive property of chisquare.
- It X & Y are two independent distribution chi square variate the with $n_1$ & $n_2$ d.f. then (X ± Y) is also a chi-square with ($n_1 \pm n_2$) d.f.
- If n is large then $\lim_{n\to\infty} xn^2 \to$N{n, 2n). This is known as the limiting property of the chi-square.
- The chi-square distribution is used to test whether a hypothetical value of the population variance is true or not.
- It is used to test a test of goodness of fit.
- It is used to test the independent of attributes.
- It is used to test the homogeneity of several population variances.
- It is used to test the equality of several population correlation coefficients

**Application of Chi-Square Distribution**

- Chi- square distribution has a large number of applications in statistics some of the most important applications are as follows.
- To test if the hypothetical value of the population variance
- To test the 'goodness of fit'
- To test the independence of attributes.
- To test the homogeneity of independent estimate of the population variance.
- To combine various probabilities obtained from independent experiments to given a single test of significance

**Student t-distribution**

Let $x_1$, $x_2$, $x_3$ …..$x_n$ be a random sample size of size and drawn from a normal population with mean μ variance $\sigma^2$. Then student t-distribution is defined as

$t = \frac{\bar{x}-\mu}{s\sqrt{n}}$

Or $t = \frac{\bar{x}-\mu}{s\sqrt{n-1}}$ Follows student t-distribution with (n-1) degree of freedom and where

$\bar{x} = \frac{\Sigma x}{n}$ and $S^2 = \frac{\sum(x-\bar{x})^2}{n-1}$

**Properties of t-distribution**

The important properties of t-distribution are given as follow:

a) Like Z-distribution, t-distribution is a continuous distribution having symmetrical and bell shaped curve. The value of 't' ranges from. - ∞ to ∞ i.e. - ∞ < t < ∞
b) We use the notation $t \sim t_n$ to show that the statistics 't' follows student's t-distribution with n degrees offreedom.
c) As n→∞ the t-distribution tends to standard normal distribution.
d) Mean of the t-distribution is zero and variance is equal to $\frac{v}{v-z}; v > 2$
e) The t-distribution is symmetrical i.e.$\beta_1 = 0 \Rightarrow \lambda_1 = 0$.
f) For t-distribution P2 = 3 (~ ~ ~). for large degrees of freedom p2 = 3, hence for large degrees of freedom t-distribution tends to normal distribution.
g) The t-distribution curve is unimodal with mean = mode = median = 0.
h) The t-axis is the asymptote of the t-distribution.



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i) The t-distribution is flatter than the normal distribution and there is a different t-distribution for every possible sample size. For example a t-distribution for a sample size of 15 is different than the t-distribution for a sample size of 2 and so on.

**Assumption about t-Distribution**

For derivation of student's t-distribution following basic assumption are made

**a)** The parent population from which the samples are drawn is normal with mean $\mu$ and variance $\sigma^2$
**b)** All observations in the sample are independent i.e. one item selected in the sample doesn't effect the other items included in the sample.
**c)** The sample size is small i. e. less than 30 as a usual practice. Also the sample should not certain less tan 5 observations.
**d)** The hypothetical value $\mu_0$ of $\mu$ is a correct value of population mean of parent population form which the sample are drawn.
**e)** The sample values are correctly measured and recorded.
**f)** The value of population standard deviation $\sigma$ is unknown.

**Application of t-Distribution**

The student's t-distribution has wide number of application in statistics, same of which are enumerated below

**a)** To test it the sample mean ($\bar{x}$) differ significantly from the hypothetical value $\mu$ of the population mean, Population variance being unknown.
**b)** To test the significance of the difference between two independent mean, The population variance being equal but unknown.
**c)** Paired t-test for difference of two means.
**d)** Test of significance of an observed sample correlation coefficient
**e)** Test of significance about the slope parameter $\beta_0$ of regression equation
**f)** Test of significance about the regression coefficient $\beta_{xy}$

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CHAPTER SEVEN

# ESTIMATION

**Parameter and statistics**

a) **Parameter:** The statistical constants of whole population are called parameter such as population mean (μ), population variance ($\sigma^2$). population proportion (P). Population correlation coefficient (p), population regression coefficient ($\beta_{vx}$) etc. out of various parameters mean and variance are largely used.

b) **Statistics:** According to Ronald A. Fisher the statistical constants of the sample selected from the populat1on are called statistics; in general statistics are the function of observable random variable and doesn't involve any unknown parameters.

| Sample statistic | Population parameter |
|---|---|
| Samle mean ($\bar{x}$) | μ |
| Standard deviation (ρ) | σ |
| Proportion (p) | P |
| Correlation coefficient (r) | ρ |

$\mu = \bar{x} \pm SE(\bar{x})$
$= \bar{x} \pm \frac{\sigma}{\sqrt{n}}$
$= \bar{x} \pm \frac{\rho}{\sqrt{n}}$
Where $\bar{x}$=mean; SE=standard error; ρ= $\sigma$

**Standard error and sampling Distribution**

The standard deviation of the sampling distribution of sample statistics is known as **standard error (S.E.) of the statistics**

| Sample | Standard error |
|---|---|
| 1. $\bar{x}$ | 1. $\frac{\sigma}{\sqrt{n}} Z_{\frac{a}{2}}$ or $\frac{s}{\sqrt{n}} Z_{\frac{a}{2}}$ or S.E($\bar{x}$) |
| 2. $\bar{x}_1 - \bar{x}_2$ | 2. S.E($\bar{x}_1 - \bar{x}_2$) = $(\sqrt{\frac{\sigma_1}{n_1} + \frac{\sigma_2}{n_2}}) Z_{\frac{a}{2}}$ |
| 3. p | 3. $S.E(p) = (\sqrt{\frac{pq}{n}}) Z_{\frac{a}{2}}$ |
| 4. $p_1 - p_2$ | 4. $S.E(p_1 - p_2) = \sqrt{\frac{p_1q_1}{n_1} + \frac{p_2q_2}{n_2}}$ |

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**Estimation:** It is the process of estimate the population parameter on the basis of sample statistic with desire degree of precision . The main objective of estimation is to obtain guess or estimate unknown.

**Point estimation:** It is the process of estimating population parameter in the form of certain interval through sampling statistic. For example a sample mean ( $\bar{x}$ ) and a sample variance $s^2$ is the point estimate of population mean μ and population variance $\sigma^2$. We write as point estimator of **μ** is $\bar{x}$ as $\hat{\mu} = \bar{x}$ and that of $\sigma^2$ a s $\hat{\sigma} = s^2$

**Interval of estimation:** It is the process of estimating population parameter in the form of certain interval through sampling statistic. There are two ways for indicating error by the extent of its range and by the probability of the true population parameter lying within that range. Example:
$\mu = \bar{x} \pm \frac{\sigma}{\sqrt{n}}$
$P = p \pm \frac{\sigma}{\sqrt{n}}$
**sometimes, interval of estimation= confidence interval**



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**Properties of good estimator**

A good estimator is one which is as close to the true value of the parameter as possible. The following are the some of the criteria that should be satisfied by a good estimator.

a. **Unbiasedness**: An estimator or statistics In which is a function of the asmple observations $x_1$, $X_2$ . ... $X_n$ is said to be an unbiased estimator of the corresponding population parameter θ if $E(t_n) = \theta$

b. **Consistency:** statistics $t_n$ is said to be consistent estimator of the population parameter θ if as the sample size increase the value of the sample statistics become very close to the value of the population parameter.

c. **Sufficiency:** A statistics $t_n$ based on sample of n observation is said to be sufficient for a parameter θ if it contains all the information in the sample regarding the parameter.

d. **Efficiency:** An unbiased estimator $t_n$, based on sample of n observation is said to be efficient estimator than any other unbiased estimator $t_n$, of population parameter θ if and only if $Var(t_n) < Var(t^o_n)$. Also the relative efficiency of $t_n$ as compared to $t^o_n$ is given as $RE = \frac{var(t_n)}{var t^o_n}$

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## CHAPTER EIGHT
## STATICAL INFERENCE

**Hypothesis :** A quantitative statement about the population parameter is called a hypothesis. A statistical hypothesis is a statement about the distribution of one or more variables

**Types of Hypothesis**

A. **Null Hypothesis :** It is the statement about population parameter which is usually a hypothesis of no difference. It is denoted by $H_0$. Null hypothesis is the hypothesis which is tested for possible rejection under the assumption that it is true**.**

B. **Alternative hypothesis.:** It is complementary of null hypothesis. It is against null hypothesis. In other word alternative hypothesis is a statement about the population parameter or parameters which provides an alternative to the null hypothesis within the range of pertinent value of parameter. It is denoted by $H_1$. It may be one tail or two tail.

**Steps of testing of hypothesis/some comments on selecting a test procedure**

1. **Set up null hypothesis $H_o$.**e.g there is no significance difference between sample statistics and population parameters.
2. **Set up alternative hypothesis($H_1$)**: e.g there is significance difference between sample statistics and population parameters. It may be one tailed or two tailed.
3. **Choose Level of significance(α):**generally it is taken 5%
4. **Test statistic:** Choose appropriate test statistics under null hypothesis $H_0$ such as t-test, z -test , f-test etc

   Test statistic $=\frac{Sample\ statistic - population\ parameter}{Standard\ error\ of\ sample\ statistic}$
5. **Critical value** : It is obtained from respective table under pre-determined value of α.
6. **Conclusion**:
   a. If calculated value is < tabulated value $H_0$ is accepted.
   b. If calculated value is > tabulated value $H_0$ is rejected and $H_1$ is accepted.

**Error in testing of hypothesis**

While testing of statistical hypothesis ,in practice there are four possibilities of making decisioneither to accept or reject. They are:

a. Accepting the null hypothesis when the null hypothesis is true.
b. Accepting the null hypothesis when the null hypothesis is false.
c. Rejecting the null hypothesis when the null hypothesis is false.
d. Rejecting the null hypothesis when the null hypothesis is true.

Here we commit two type of errors. They are:

A. **Type I error:** The error committed in rejecting null hypothesis when it is true is called type I error . It is also known as producers risk. Type I error amounts to rejecting a lot when it is good.

B. **Type II error:** The error committed in accepting null hypothesis when it is wrong is called type II error . It is also known as consumers risk. Type II error amounts to accepting a lot when it is bad.

**Level of significance** :The level of significance may be defined as the probability of type of I error which we are ready to tolerate in making a decision about $H_o$.

i.e. P (reject $H_o$ when $H_o$ is true)= α

It is our endeavour to carry out a test which minimizes both type of error; unfortunately for given set of observations, both the errors can't be controlled simultaneously. Hence it is a general practice to assign a bound to type I error and to minimize type II error, Thus one choose a value of α lying between 0 and 1 which is known as the level of significance.

**Degree of freedom:** The number of independent observation in a set is called degrees of freedom. Inother word it may be defined as the number of observation in a set minus the number of restrictions improved on it. . It is denoted by v(nu) or d.f.



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**One tailed or two tailed test:**

If an alternative hypothesis is such that it leads to one sided alternatives to the null hypothesis, it is said to be **a one tailed test**. For example, Test $H_o$: $\mu = 20$ verses $H_1$:$\mu > 20$ or $H_1$:$\mu < 20$ , the critical region or area 'A' lies only on one sided tail. Here Area 'A' lies on right tail given when $\mu > 20$ and lies in left tail given when $\mu < 20$.

If an alternative hypothesis is such that it leads to two alternatives to the null hypothesis, it is said to be **a two tailed test**. For example, Test $H_o$: $\mu = 20$ verses $H_1$:$\mu \neq 20$ leads to two sided test as $\mu$ can be greater than 20 or less than 20. In this situation half of the area of critical region lies on the left tail and the half on the right. If A is the area of the critical region $\frac{A}{2}$ is the area on both the tails

**Large sample test(z – test)**

Large sample test is generally used when the sample size is greater than 30. i.e $n \geq 30$.

**Assumptions**

a.In this test the sample drawn from population are normally distributed.
b.Population standard deviation is known
c. Samples are independent

**Test of significance (For mean)**

**A. <u>For single mean</u>**

a. **Set up null hypothesis $H_o$:** Ho:.$\mu = \mu_o$ or $H_0$:$\mu = \bar{X}$
b. **Set up alternative hypothesis($H_1$) :** $H_1$: .$\mu \neq \mu_o$ or $H_1$:$\mu \neq \bar{X}$
c. **Choose Level of significance($\alpha$):** Determine the level of significance at which hypothesis is to betested.
d. **Test statistic:** Under null hypothesis Ho:.$\mu = \mu_o$ the statistic is given by
**Test statistic (Z)**=$\frac{\bar{x}-\mu}{S.E(\bar{x})} = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{\bar{x}-\mu}{s/\sqrt{n}}$ where $\mu$=given mean and $\sigma$ is given S.D

$\mu = \bar{x} \pm \frac{\sigma}{\sqrt{n}} Z_{\frac{\alpha}{2}} = \bar{x} \pm \frac{s}{\sqrt{n}} Z_{\frac{\alpha}{2}}$

e. **Critical value** : Find the critical value of Z as $Z_\alpha$ from table
f. **Conclusion**:
i. If calculated value $Z < Z_\alpha$ , $H_0$ is accepted.
ii. If calculated value is $Z > Z_\alpha$ , $H_0$ is rejected and $H_1$ is accepted.

**B. <u>Difference of Two mean</u>**

Process same as in single mean except value Z

$Z = \frac{\bar{x}-\bar{x}}{S.E(\bar{x}-\bar{x})} = \frac{\bar{x}-\bar{x}}{\sqrt{\frac{s^2}{n_1}+\frac{s^2}{n_2}}} = \frac{\bar{x}-\bar{x}}{\sqrt{\frac{\sigma^2}{n_1}+\frac{\sigma^2}{n_2}}}$

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**Test of significance for Population proportion**

**A .<u>single Proportion</u>**

a. **Set up null hypothesis $H_o$:** Ho:$P = P_o$
b. **Set up alternative hypothesis($H_1$) :**
$H_1$: .$P \neq P_o$ **(**For two tailed**)**
$H_1$: .$P < P_o$ **(**For one left tailed**)**
$H_1$: .$P > P_o$ **(**For one right tailed**)**
c. **Choose Level of significance($\alpha$):** Determine the level of significance at which hypothesis is to be tested.
d. **Test statistic:** Under null hypothesis Ho:.$\mu = \mu_o$ the statistic is given by
**Test statistic (Z)**=$\frac{p-P}{S.E(p)} = \frac{p-P}{\sqrt{\frac{PQ}{n}}} = \frac{p-P}{\sqrt{\frac{PQ}{n}}}$
where P= Proportion of claimed value and p sample proportion
**Q=1 – P** and n= no. of sample


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$\mathbf{P = P \pm Z_{\frac{\alpha}{2}} S.E = p \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{PQ}{n}} = p \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{PQ}{n}}}$ is confidence interval of population interval.

**q=1 – p**

e. **Critical value** : Find the critical Z as $Z_\alpha$ from table

f. **Conclusion**:

i. If calculated value $Z < Z_\alpha$ , $H_0$ is accepted.

ii. If calculated value is $Z > Z_\alpha$ , $H_0$ is rejected and $H_1$ is accepted.

**B. <u>Two proportion</u>**

If $p_1$ and $n_1$ are proportion and number of samples drawn fron 1st population $p_2$ and $n_2$ are proportion and number of samples drawn fron 2nd population then test statistic is given by:

$$Z = \frac{p_1 - p_2}{S.E(p_1 - p_2)} = \frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

**Small sample test(t-test )**

**Assumptions for t-test**

a. The sample size is less than 30
b. The sample drawn from the population are normally distributed
c. Population varience or S.D is not known
d. Sample are independent and one sample does not effect other sample
e. The sample values are correctly measured and correct

**T-test for Significance difference of single mean**

a. **Set up null hypothesis $H_0$:** Ho:. $\mu = \mu_o$
b. **Set up alternative hypothesis($H_1$) :**
   $H_1$: . $\mu \neq \mu_o$ **(**For two tailed**)**
   $H_1$: . $\mu < \mu_o$ **(**For one left tailed**)**
   $H_1$: . $\mu > \mu_o$ **(**For one right tailed**)**
c. **Choose Level of significance(α):** Determine the level of significance at which hypothesis is to betested.
d. **Test statistic:** Under null hypothesis Ho:.$\mu = \mu_o$ the statistic is given by

   **Test statistic (t)** $= \frac{\bar{x} - \mu 0}{\frac{S}{\sqrt{n}}}$

   where $\bar{x} = \frac{\Sigma x}{n}$ is sample mean $\mu_0$ =claimed value of population mean.

   $s^2 = \frac{\Sigma (X - \bar{x})^2}{n-1} = \frac{\Sigma x^2}{n-1} - \frac{(\Sigma x)^2}{n(n-1)}$ **or** $s^2 = \frac{\Sigma (X - \bar{x})^2}{n} = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$

e. **Degree of freedom = n – 1**
f. **Critical value** : Find the critical value t as $t_\alpha$ from table

g. **Conclusion**:

i. If calculated value $t < t_\alpha$ , $H_0$ is accepted.

ii. If calculated value is $t > t_\alpha$ , $H_0$ is rejected and $H_1$ is accepted.

**T-test for Significance difference of two mean**

a. **Set up null hypothesis $H_0$:** Ho:. $\mu_1 = \mu_2$
b. **Set up alternative hypothesis($H_1$) :**
   $H_1$: . $\mu_1 \neq \mu_2$ **(**For two tailed**)**
   $H_1$: . $\mu_1 < \mu_2$ **(**For one left tailed**)**
   $H_1$: . $\mu_1 > \mu_2$ **(**For one right tailed**)**
c. **Choose Level of significance(α):** Determine the level of significance at which hypothesis is to betested.



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d. **Test statistic:** Under null hypothesis Ho:. $\mu_1 = \mu_2$ the statistic is given by
**Test statistic (t)**= $\frac{\bar{x}1-\bar{x}2}{S.E(\bar{x}_1-\bar{x}_2)} = \frac{\bar{x}_1-\bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}}$
where $s_p^2 = \frac{(n_1-1)s^2+(n_2-1)s^2}{n_1+n_2-2}$
e. **Degree of freedom = $n_1 + n_2 - 2$**
f. **Critical value** : Find the critical value t as $t_\alpha$ from table
g. **Conclusion**:
- If calculated value $t < t_\alpha$ , $H_0$ is accepted.
- If calculated value is $t > t_\alpha$ , $H_0$ is rejected and $H_1$ is accepted.

**Paired t-test**

It is concerned with the difference between the pair of related observations instead of the value of the individual observation. Paired t-test for difference of means can be applied under the following situations.

- The sample sizes are equal i.e. $n_1 = n_2 = n$ (say).
- The two samples are not independent but the sample observations are paired together i.e. the pair observations $(x_i , y_i)$ ; i = 1, 2 ... n corresponds to the same sample unit.
- The same set of sample is treating twice on the same subject matter.

The following are the steps in testing paired t-test for difference of mean or **t-test for paired** wise dependent samples is as follows:

a. **Set up null hypothesis $H_0$:** Ho:. $\mu_1 = \mu_2$ **;** there is no significance difference in the observation before the treatment
b. **Set up alternative hypothesis($H_1$) :** There is significance difference in the observation before the treatment
$H_1$: . $\mu_1 \neq \mu_2$ **(**For two tailed**)**
$H_1$: . $\mu_1 < \mu_2$ **(**For one left tailed**)**
$H_1$: . $\mu_1 > \mu_2$ **(**For one right tailed**)**
c. **Choose Level of significance(α):** Determine the level of significance at which hypothesis is to betested.
d. **Test statistic:** Under null hypothesis Ho:.$\mu_1 = \mu_2$ the statistic is given by
**Test statistic (t)** = $\frac{\bar{d}}{\frac{S_d}{\sqrt{n}}}$ , follows the student t=distribution with (n – 1) degree of freedom
where $\bar{d} = \frac{\Sigma d}{n}$ ; d= x – y
$sd^2 = \frac{\Sigma(d)^2}{n-1} - \frac{(\Sigma d)^2}{n(n-1)}$
e. **Critical value** : For α level of significance and (n – 1) degrees of freedom find the critical value t as $t_\alpha$ from table
f. **Conclusion**:
- If calculated value $t < t_\alpha$ , $H_0$ is accepted.
- If calculated value is $t > t_\alpha$ , $H_0$ is rejected and $H_1$ is accepted.

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## CHAPTER NINE
## THE ANALYSIS OF CATEGORICAL DATA

**Chi-square test** : Chi-square test ($\psi 2$) is a non- parametric test It depends only on the set of observed and expected frequencies and degree of freedom. Since chi-square test doesn't make any assumption about population parameter ,it is also called distribution free test. Chi-square test is a test which describes the magnitude of difference between observed frequency and theoretical or expected frequencies under certainassumptions.

$\psi 2=\sum \frac{O-E^2}{E}$ where E=Expected frequency and O=observed frequency

**Application of Chi-Square Distribution** Chi- square distribution has a large number of applications in statistics some of the most important applications are as follows.

A. To test if the hypothetical value of the population variance
B. To test the 'goodness of fit'
C. To test the independence of attributes
D. To test the homogeneity of independent estimate of the population variance.
E. To combine various probabilities obtained from independent experiments to given a single test of significance

**Test procedure for a population Varience**

a. **Set up null hypothesis ($H_o$):** $H_o:\sigma^2 = \sigma_0^2$; sample are drawn from the normal population with specified variance $\sigma_0^2$
b. **Set up alternative hypothesis($H_1$) :** $\sigma^2 \neq \sigma_0^2 i.e$ sample are drawn from the normal population with specified variance $\sigma_0^2$
c. **Choose Level of significance(α):** Determine the level of significance at which hypothesis is to be tested.
d. **Test statistic:** Under null hypothesis $H_o:\sigma^2 = \sigma_0^2$ the statistic is given by
**Test statistic (t)** = $\psi^2 = \sum \frac{\bar{x}-\bar{x}^2}{\sigma_0^2} = \sum \frac{ns^2}{\sigma_0^2}$, follows chi-square test distribution with (n – 1) degree of freedom
where $sd^2 = \frac{1}{n}\sum(x-\bar{x})$

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e. **Critical value** : For α level of significance and (n – 1) degrees of freedom find the critical value t as $t_\alpha$ from table
f. **Conclusion**:

- If calculated value $t < t_\alpha$ , $H_0$ is accepted.
- If calculated value is $t > t_\alpha$ , $H_0$ is rejected and $H_1$ is accepted

**Chi-square test test of goodness of fit**

Following are the steps for test of goodness of fit

a. **Set up null hypothesis ($H_o$):** The fitting of certain distribution is good
b. **Set up alternative hypothesis($H_1$) :** The fitting of certain distribution is not good
c. **Choose Level of significance(α):** Determine the level of significance at which hypothesis is to be tested.
d. **Test statistic:** Under null hypothesis Ho the statistic is given by
**Test statistic (t)** = $\psi^2 = \sum \frac{O-E^2}{E}$, follows chi-square test distribution with (n – 1) degree of freedom
where $sd^2 = \frac{1}{n}\sum(x - \bar{x})$
e. **Critical value** : For α level of significance and (n – 1) degrees of freedom find the critical value t as $t_\alpha$ from table
f. **Conclusion**:

- If calculated value $t < t_\alpha$ , $H_0$ is accepted.
- If calculated value is $t > t_\alpha$ , $H_0$ is rejected and $H_1$ is accepted

**Two way contigency table and test of independence of attributes**: A contingency table is a two-way table in which the columns are classified according to one criterion or attribute and rows are classified according to the other criteria or attribute. Each cell contains that the number of items $O_{ij}$ possessing the quality of $i^{th}$ row and $j^{th}$ column. Where i = 1, 2, ... nand j = 1, 2, ... m. In such a case the contingency tabl is said to be of order (n x m). Contingency table helps to test the independence of two attribute. A two ways contingency table is as

**Test of independence of attributes** In order to test whether attribute or characteristic A and B are independent i.e. there is no association or relationship between the two characteristics or attribute A and B. we use the following step for of independent of attributes. Similarly to the **TEST OF HOMOGENEITY** oftwo or more then two population we use the same process with slightly different hypothesis.

a) **Set up null hypothesis** $H_o$: Two attributes (two categorical variable) A and B are independence i.e. there is no relationship or association between them
b) **Set up alternative hypothesis $H_1$**: Two attributes (two categorical variable) A and B are dependence i.e. there is some relationship or association between them.
c) **Fix the level of significance** α
d) **Test statistics**: Under null hypothesis Ho. the test statistics is
$= \psi^2 = \sum \frac{O-E^2}{E}$ follows chi-square distribution with (r-1) (c-1) ; where,
O=observed frequency, E= expected frequency
r= number of rows, c=numbers of columns.
The expected frequency in a cell is the product of its row total and column total divided by the total frequency, thus $e = \frac{RT \times CT}{N}$ where RT=row total, CT- column total and N is the grand total.
e) **Critical Value**: For a-level of significance and (r-1) (c-1) degrees of freedom, find the critical value of ψ2 from table as $\varphi^2_\alpha$
f) **Conclusion**: if calculated value of ψ2 is less than tabulated value we accept our null hypothesis and otherwise reject it.

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